



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
SCIENCE AND TECHNOLOGY DEVELOPMENT**

HIGHER EDUCATION EXAMINATIONS COUNCIL

SECTION A

QUESTION 1

- a) Express 492.731_{10} in Octal form (2 marks)
- b) Round off 0.625 to 2 d.p (1 mark)
- c) Transpose $V = \frac{\pi h(3R^2 + h^2)}{6}$ to make R the subject (3 marks)
- d) Solve by completing the Square $2x^2 - 4x - 3 = 0$ (4 marks)
- e) Using the factor theorem show that when $x^3 + 2x^2 + 3x + 4$ is divided by $x + 2$ the remainder is 2. (3 marks)
- f) Simply $\log 64 - \log 128 + \log 32$ (2 marks)
- g) Determine the total surface area of a Cone of radius 5cm and perpendicular height 12cm. (3 marks)
- h) Divide $x^3 + 5x^2 + 10x + 15$ by $x + 2$ (4 marks)
- i) Solve the two Simultaneous equations using the matrices method
 $9x + 3y = 39$
 $4x + 2y = 24$ (5 marks)
- j) Determine $\int 3\sqrt{x} dx$ (2 marks)
- k) Find the derivative of $y = \frac{1}{x^2} + \sqrt[3]{x}$ (2 marks)
- l) Find the coordinates of the points on the curve $y = x^2$ given that $x = 4$ (2 marks)
- m) Express 222_{10} in binary form (2 marks)
- n) Factorise $x^4 - 16$ completely (2 marks)
- o) Simply $\frac{7^{-2} \times 3^4}{3^{-2} \times 7^5 \times 5^{-2}}$ (3 marks)

PAPER NO: 346/13/S08 -ENGINEERING MATHEMATICS

QUESTION 8

Differentiate $y = x^3 + 3$ with respect to x from first principle

(12 marks)

...../km



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR MECHANICS, AUTO - ELECTRICS, DPF, PRECISION MACHINING,
MOTOR CYCLE MECHANICS, MVBR**

SUBJECT: Engineering Mathematics

PAPER NO: 346/13/S08

DURATION: 3 Hours

OCTOBER/NOVEMBER 2015 EXAMINATION

REQUIREMENTS

Scientific calculator

INSTRUCTIONS TO CANDIDATE

- 1. Section A: Answer all Questions**
 - 2. Section B: Answer any five (5) questions**
-

PAPER NO: 346/SO8 – ENGINEERING MATHEMATICS

SECTION A

QUESTION 1

- a) Find the length of a straight line which passes through points:

A (2:-3)

B (5:4)

(2 marks)

- b) From the first principle solve

$$(2x^2 + 4) \frac{\delta y}{\delta x}$$

(6 marks)

- c) $\int (3x^3 - 4x^2 + x) dx$

(3 marks)

QUESTION 2

- a) Solve the equation:

$$0.25 x^2 + 1.25x + 1.5 = 0$$

(3 marks)

- b) Simplify:

$$3\log + \log 4 + 2 \log 3 - \log 18$$

(3 marks)

- c) Find the inverse of matrix M:

$$\begin{pmatrix} 7 & 5 \\ 2 & 3 \end{pmatrix}$$

(3 marks)

QUESTION 3

- a) Rewrite in the nested form the following expression and then evaluate

$$f(x) = x^5 - 3x^4 + 4x^3 + 7x^2 + 2x + 8$$

$$\text{When } x = -1$$

(2 marks)

- b) i) Transpose for x ; $y = ax^n$

(3 marks)

- ii) Find $(2x + 3)^2$

(2 marks)

- c) Write down the formula for calculating the area of a flat circular ring (annulus)

(2 marks)

- d) Evaluate, $\frac{2^2 \times 2^3}{24}$

(3 marks)

- e) A single cylinder engine has a connecting rod 120 mm long and stroke of 60 mm. Determine the distance moved by the piston from TDC when the

PAPER NO: 346/SO8 – ENGINEERING MATHEMATICS

connecting rod reaches its maximum angularity relative to the cylinder axis (5 marks)

- f) Express 48_{10} to base 2 (binary) (3 marks)

SECTION B

QUESTION 4

- a) Factorise: $x^3 - 2x^2 - 5x + 6$ (6 marks)

- b) Solve by completing the square the following equation

$$x^2 + 3x - 4 = 0$$

(6 marks)

QUESTION 5

- a) A petrol tank 0.5m long has spherical ends of radius 0.3m. Determine:

PAPER NO: 346/SO8 – ENGINEERING MATHEMATICS

- b) Find the equation of the tangent to $y = x^2$ at the point (1:1) and the equation of the tangent to $y = \frac{1x^3}{6}$ at the point $(2; \frac{4}{3})$. Show that the tangents are parallel. (6 marks)

QUESTION 9

- a) Solve the indicial equation

$$7(14.3^{x+5}) \times 6.4^x = 294$$

Correct of 4 significant figures

(6 marks)

- b) Prove without using logs

$$\frac{\sqrt{27} + \log \sqrt{8} - \log \sqrt{125}}{\log 6 - \log 5} = \frac{3}{2}$$

(6 marks)

QUESTION 10

- a) A certain car has a wheel base of 4m and wheel track of 1.6m. When the car is turning on a certain radius the steering angle of the inner wheel is 22°

Calculate

- The angle of the outer wheel
 - The length of the radius (measured to the centre of the rear axle)
- (6 marks)

- b) The starting point of a hill climb course has an angle of depression of 12° when viewed from the finishing line which is 100 m above the level of starting point

- Find the distance of the course assuming that it is 12% longer than the gradient.
- (6 marks)

...../dn

SECTION A

QUESTION 1

- a) What is the difference between speed and velocity? (1 mark)
- b) State Newton's Third Law of motion. (1 mark)
- c) What is a machine? (1 mark)
- d) How many radians are there in 360° ? (1 mark)
- e) Define weight (1 mark)
- f) Write down the formula for calculating potential energy (1 mark)
- g) What is specific heat capacity? (1 mark)
- h) A pressure of 1 bar =KPa (1 mark)
- i) Define the term density (1 mark)
- j) Divide 35m into the ratio 2:5 (1 mark)

SECTION B

QUESTION 2

- a) The tooth numbers in a four-speed gearbox are arranged to be interchangeable so that only four types of gear wheel are required. These tooth numbers are 20, 26, 33 and 39. With the aid of sketch diagrams, calculate the gear ratio in 2nd gear and 3rd gear. (5 marks)
- b) The reading on a compression gauge is 690 kN/m^2 . Express this reading as an absolute pressure in kN/m^2 . Take atmospheric pressure is 101.3 kPa . (3 marks)
- c) Temperature of air compressed in an engine is 368 K . Express this figure in degrees Celsius. (3 marks)
- d) A quantity of a gas occupies a volume of 5 m^3 at a pressure of 140 kPa absolute. It is then compressed at constant temperature until the new pressure is 560 kPa absolute. Determine the final volume of the gas. (3 marks)
- e) A vehicle is accelerated uniformly at 1.5 m/s^2 from a speed of 27 km/h . Calculate the time it takes to attain a speed of 81 km/h . (5 marks)

PAPER NO: 346/SO9 – AUTOMOTIVE ENGINEERING SCIENCE

- f) A morse test was carried on a four cylinder engine. The following data was collected:

A cylinder firing brake load	=	141.6N
No. 1 cylinder cut out	=	100N
No 2 cylinder cut out	=	104N
No. 3 cylinder cut out	=	101.2N
No 4. Cylinder cut out	=	103,2N

- Calculate indicated power (5 marks)
- g) State CHARLES' LAW (3 marks)
- h) State the formula to calculate power transmitted by belts (3 marks)
- i) With a velocity ratio of 6, a machine lifts a mass of 900kg using just an effort of 2000N. Calculate:
- a) the efficiency of the machine
- b) the effort required to overcome friction (10 marks)

SECTION C

QUESTION 3

At a given instant, a vehicle of mass 1500kg has a speed of 18km/h when it is subjected to a uniform acceleration of 2m/s^2 until its speed is 72km/h. Determine:

- a) the time taken to increase the speed
- b) the distance travelled during the acceleration
- c) the accelerating force (10 marks)

QUESTION 4

A single plate clutch of 0.15m effective diameter is lined with material of coefficient of friction 0.35. If the total spring force is 2.5kN. Calculate:

- a) the torque transmitted (5 marks)
- b) the power transmitted at 3000 rev/min (5 marks)

QUESTION 5

A vehicle has a mass of 1325kg and the coefficient of friction between the tyres and around is 0.4. Determine the actual retarding force which can be used to stop it without causing it to skid? If the actual retarding force at the ground is 0.75 of the maximum, and is constant, calculate the work done in bringing the vehicle to rest in a distance of 20m.(10 marks)

QUESTION 6

The engine is suspended 5m above the ground and has a mass of 200kg.

- a) Calculate the potential energy of the engine due to its position above the ground.
- b) Due to a fault the engine falls freely to the ground from that height, determine the velocity and the kinetic energy of the engine at the point of impact with the ground.
- c) Determine the KE and the PE of the engine after falling 4m.

(10 marks)

QUESTION 7

A bus, 5.4m long, consists of two side members and a number of cross members. Each side member can be considered as a beam, simply supported at two points A and B, 3.6m apart. A beam positioned 0.9m from the frontend of the frame and subjected to the following concentrated loads, engine support (front) 2kN, engine support (rear) 2.5kN, gearbox support 0.5kN and body WkN. The distances of these loads from the end of the frame are respectively 0.6m, 1.8m, 2.4m and 3m. If the reaction at A is 8.5kN determine.

- a) The magnitude of the load W due to the vehicle body
- b) The magnitude of the support reaction at B

(10 marks)

QUESTION 8

A piece of copper of mass 1.5kg is heated to 140°C and quickly placed in a copper vessel of mass 0.75kg and containing 0.75kg of water both at a temperature of 10°C. The temperature of the copper vessel and its contents are thus increased to 30°C. What is the specific heat capacity of the copper? Take the specific heat capacity of water as 4200J/kgK.

(10 marks)

QUESTION 9

A steel brake rod is 1.2m long and is subjected to a maximum load of 4.5kN. If the extension of the rod is not to exceed 0.384mm, determine a suitable diameter for the rod. Take Young's modulus $E = 200\text{GPa}$.

(10 marks)

...../ab



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SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR MECHANICS, DPF; AUTO-ELECTRICS; MVBR; PRECISION
MACHINING; MOTOR CYCLE MECHANICS**

SUBJECT: Engineering Science

PAPER NO: 346/13/S09

DURATION: 3 Hours

OCTOBER/NOVEMBER 2015 EXAMINATION

REQUIREMENTS

Scientific calculator, Graph paper, Drawing instruments

INSTRUCTIONS TO CANDIDATE

SECTION A: Answer ALL questions. (40 marks)

SECTION B: Answer any FIVE (5) questions. (60 marks)

✓ QUESTION 5

- a) Tin has a melting point of 232°C and specific heat capacity of $0,23\text{KJ/kgK}$ and a latent heat of fusion at $60\,000\text{ J/kg}$.
- i) What quantity of heat is required to melt 20kg of tin originally at 32°C ? (6 marks)
- ii) Steel has specific heat capacity of $0,48\text{KJ/kg}$. If $1,5\text{kg}$ of steel at a temperature of 32°C are plunged into molten tin referred to a (i) above. What mass of tin solidifies when the steel acquires a temperature of 232°C ? (6 marks)

2 ✓ QUESTION 6

A disc brake assembly has a pair of opposing friction pads pressing on it at an effective radius of $0,12\text{m}$. The contact area of each pad is $0,02\text{m}^2$ and the coefficient of friction between pads is 0.3 .

If the hydraulic pressure in the brake line is 700KPa and the disc speed reduces uniformly from 10rev/min to 6 rev/min . Determine:

- a) The braking torque (4 marks)
- b) Power absorbed by the brakes of the wheels fitted with 4 such brake assembling. (4 marks)
- c) Total heat generated at the brakes (4 marks)

7 ✓ QUESTION 7

A single cylinder 2 stroke diesel engine develops $6,71\text{KW}$ at 2800 rev/min .

- a) Calculate the work done on the piston in joules in one stroke. (6 marks)
- b) If the piston is 100mm diameter and the stroke is 150mm , calculate the average pressure on the piston in KN/m^2 . (6 marks)

✓ QUESTION 8

In an experiment to verify the principle of moments, a horizontal uniform beam 2m long was supported at each end and a load of 30N was placed $0,5\text{m}$ from the left hand support. The reaction at the right hand support was found to be $12,5\text{N}$. Find:

- a) The mass of the beam (6 marks)
- b) The reaction at the left hand support (6 marks)

The engine has a mass of 100 kg and is suspended from a crane by a single cable. The engine is 4 m above ground level.

- a) Determine the potential energy of the engine due to its position above the ground. (2 marks)

- b) Due to a fault in the engine it falls freely to the ground from that height. Calculate the velocity and the kinetic energy of the engine at the point of impact with the ground. (5 marks)

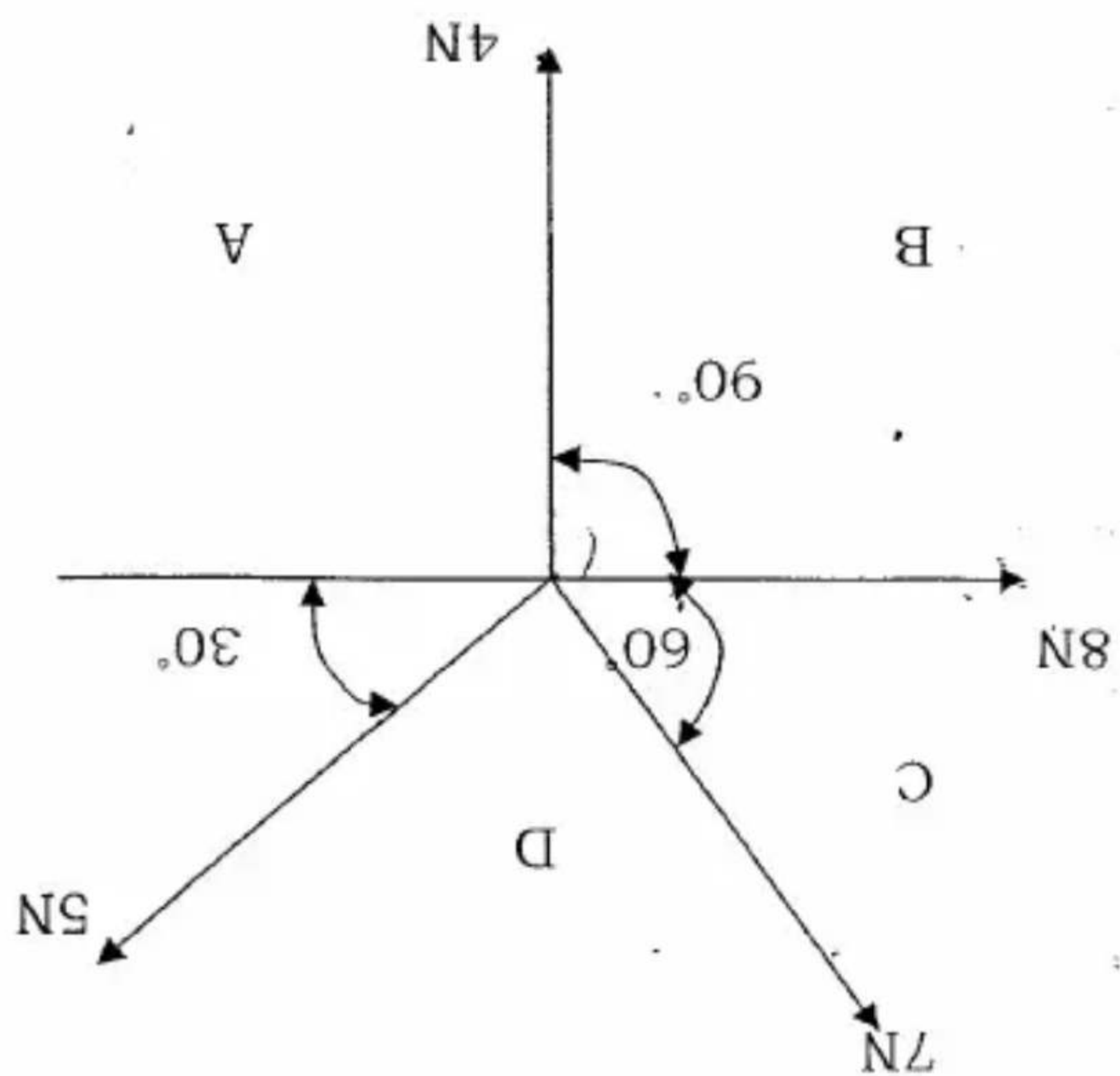
- c) Determine the kinetic energy and the potential energy of the engine after falling 3 m. (5 marks)

Neglect air resistance.

(5 marks)

QUESTION 10

Four coplanar forces act at a point as shown below. Determine their resultant in magnitude and directions graphically.



(12 marks)

.../ab



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SCIENCE AND TECHNOLOGY DEVELOPMENT
HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)

NATIONAL CERTIFICATE

IN

MOTOR VEHICLE MECHANICS, AUTOMOBILE ELECTRICS AND
ELECTRONICS, MOTOR CYCLE MECHANICS, VEHICLE BODY
REPAIR DIESEL PLANT, FITTING AND PRECISION MACHINING

SUBJECT: Engineering Mathematics and Science
PAPER NO: 346/S03

DURATION: 3 Hours

MARCH/APRIL 2014 EXAMINATION

REQUIREMENTS

Scientific Calculator

INSTRUCTIONS TO CANDIDATE

SECTION A: Answer ALL questions. (10 marks)
SECTION B: Answer any FIVE (5) questions. (30 marks)
SECTION C: Answer ALL questions. (10 marks)
SECTION D: Answer any FIVE (5) questions. (50 marks)

This paper consists of 5 printed pages.

SECTION A

QUESTION 1

- 1.1 Convert 0.526_{10} to octal form. (1 mark)
- 1.2 Express $21/24$ in decimal form to 2 decimal places (1 mark)
- 1.3 What percentage of 70 is 2? (1 mark)
- 1.4 $\log 2 + \log 6 - \log 4$ is what? (1 mark)
- 1.5 The volume of tank is 0.095m^3 . How many litres of petrol will be required to fill the tank? (1 mark)
- 1.6 The radius of a road curve is 150m. Find the angle turned through in 1 second in radians and degrees by a vehicle travelling at 72km/h. (2 marks)
- 1.7 Differentiate $y = x^5 - 2x^4 + 3x^3 - 4x^2 + 5$ (1 mark)
- 1.8 $\int (6x^5 + 4x^3 - 6)dx$ (1 mark)
- 1.9 $A = \begin{pmatrix} 4 & -3 \\ 1 & 2 \end{pmatrix}$ find the determinand (1 mark)

SECTION B

QUESTION 2

- ✓ 2.1 Solve the equations; $3x + 6y = 39$ and $4x - 3y = 8$ (6 marks)
- 2.2 Find the volume and capacity in litres of a tank with circular ends of diameter 0.3m and length 1.12m. (6 marks)
- ✓ 2.3 Divide $4x^3 + x^2 - 16x - 4$ by $x - 2$ (6 marks)
- ✓ 2.4 Differentiate with respect to x
 - a) $x^4 - 9x^3 + 6$ (2 marks)
 - b) $(x - 4)(x - 4)$ (2 marks)
 - c) $x^{-2}(1 + x)$ (2 marks)
- 2.5 Use matrices to solve the simultaneous equations $u = x$

$$\begin{aligned} 4x - 3y &= 18 \\ x + 2y &= -1 \end{aligned}$$

$$3x + 6y = 39 \quad (6 \text{ marks})$$

$$\begin{aligned} 3x &= 39 - 6y \\ x &= 13 - 2y \end{aligned}$$

$$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du}$$

$$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du}$$

2π

2x¹⁻

✓ 2.6 If $P = i + 3j - k$ and $q = 2i + j + 4k$

Find :

- a) $p + q$ (2 marks)
- b) $p - q$ (2 marks)
- c) $|p + q|$ (2 marks)

- ✓ 2.7 a) Integrate $\int 7x^4 dx$ (2 marks)
- b) Find the value of c in (a) above when $x = 0$ and $y = 2$ and rewrite the integral replacing c with the calculated value (4 marks)

SECTION C

QUESTION 3

- 3.1 A vehicle is accelerated uniformly at 1.5m/s^2 from 27km/h to 81km/h . Calculate the time taken. (3 marks)
- 3.2 Find the work done when a force of 5970N lifts an engine through a vertical distance of 1.2m . (1 mark) ✓
- 3.3 Given the frictional force $F = 3\text{N}$ and normal reaction $M = 10\text{N}$, find the coefficient of friction μ . (1 mark)
- 3.4 Define resultant force. (1 mark)
- 3.6 State the difference between indicated power and brake power. (1 mark)
- 3.7 Write down the formula for calculating Young's modulus of elasticity. (1 mark)
- 3.8 What is specific heat capacity? (1 mark)
- 3.9 Sketch a class 1 type of lever (1 mark)

$-2 (-2)$
 4
 $-2 (+2)$

(i) $V = u + at$
 (ii)

SECTION D

QUESTION 4

- a) A simple screw jack has a thread of lead 5mm. If the effort applied at an effective radius of 210mm, determine the velocity ratio. (4 marks)

- ✓ b) It is found that the jack can lift a load of 500N which is W, when the effort P is 20N and that W is 1000N when P is 30N.

Presume that P and W are related by a law of the type $P = aW + b$, determine when the load is 1500N,

- i) The effort (2 marks)
ii) The mechanical advantage (2 marks)
iii) The efficiency of the jack (2 marks)

QUESTION 5

An engine has a stroke of 100mm and a connecting rod length of 200mm between centres. The crank is 45° past top dead centre when the force on the piston is 5000N. By means of graphical solutions, determine:

- a) The angle between the axis of the bore and the centre line of the con rod. (2 marks)
b) The force acting along the connecting rod (3 marks)
c) The force between the thrust force of the piston and the cylinder wall. (5 marks)

QUESTION 6

A car is travelling at a uniform speed of 50km/hr. It maintains this speed for 2 minutes. It is then uniformly accelerated to 100km/h in the next one minute and then immediately retarded and brought to rest after a total of 6 minutes travelling time. Find:

- a) The distance covered whilst accelerating (2 marks)
b) The distance covered whilst braking (6 marks)
c) The total distance covered (2 marks)

$$v^2 = u^2 + 2as$$
$$V = u + at$$

QUESTION 7

A lorry hauls a trailer at 72km/h when exerting a steady pull of 800N on the tow rope. Calculate:

- a)
 - i) The work done in 20 minutes
 - ii) in megajoules and in kilowatt hours(5 marks)
- b) The power required (5 marks)

QUESTION 8

A wheel 900mm diameter, whose axis is horizontal, rotates through an angle of 130° . What is the height of a point on its rim above what was originally the lowest position. Use trigonometry. (10 marks)

QUESTION 9

8000mm³ of methyl alcohol having a coefficient of volumetric expansion of $120 \times 10^{-5}/K$ is contained in a aluminium container that is just full at $-10^\circ C$. Calculate the volume of alcohol that will overflow when its temperature rises to $40^\circ C$. Take coefficient of linear expansion of aluminium to be $24 \times 10^{-6}/K$. (10 marks)

QUESTION 10

A horizontal beam AB is 6m long and is simply supported at A and B. Loads of 30kN and 60kN are carried at points 2m and 4m respectively from A.

- a) Find the magnitude of reaction of A and B. (4 marks)
- b) If an extra load of 60kN is to be added to the beam in such a position that reactions are equal what will these reactions be at what distance from B must the 60kN force be situated? (6 marks)

QUESTION 11

A wooden tray rest on a horizontal surface and it is found that with a mass of 3kg in the tray a horizontal force of 10N is just sufficient to cause the tray to slide. When the 3kg mass is replaced by a mass of 7kg it is found that the horizontal force required to cause the tray to slide is doubled.

Calculate:

- a) The coefficient of friction between the tray (with mass) and the surface. (6 marks)
- b) The work done in moving the tray through a distance of 5m with an 11kg mass in the tray. (4 marks)

...../ab

$$F = ma$$

5

$$F_f$$

$$P = \text{Work per m}$$



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MINISTRY OF HIGHER AND TERTIARY EDUCATION

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR MECHANICS, AUTOMOBILE ELECTRIC & ELECTRONICS,
DIESEL PLANT FITTING, VEHICLE BODY REPAIR, PRECISION
MACHINING AND MOTOR CYCLE MECHANIC**

**SUBJECT: Engineering Science and
Mathematics**

PAPER NO: 346/S03

OCTOBER/NOVEMBER 2012 EXAMINATION

REQUIREMENTS

**Answer Book, Graph paper, Scientific Calculator,
Drawing Instruments**

INSTRUCTIONS TO CANDIDATE

- | | | |
|-------------------|--|-------------------|
| SECTION A: | Answer ALL questions. | (10 marks) |
| SECTION B: | Answer any FIVE (5) questions out of 7. | (30 marks) |
| SECTION C: | Answer ALL questions. | (10 marks) |
| SECTION D: | Answer any FIVE (5) questions out of 8. | (50 marks) |

This paper consists of 6 printed pages.

M. M

SECTION B

Answer any FIVE (5) questions.

QUESTION 2

2.1 Determine the remainder resulting from dividing. Use long division method $4x^3 + 13x + 33$ by $2x + 3$ (6 marks) ✓

2.2 Solve the initial equation: $3^{x+1} = 2^{2x-3}$ correct to 2 decimal places. (6 marks) ✓

2.3 The starting point of a hill-climb course has an angle of depression of 12 degrees when viewed from the finishing line which is 100 metres above the level of the starting point.

Find the distance of the course assuming that it is 12 per cent longer than the gradient. (6 marks)

2.4 a) Find the 1st derivative of $y = (\sqrt{x})^3 + \frac{1}{(\sqrt[3]{x})^5} + \frac{2x^7}{x^4} + 2$ (3 marks)



Differentiate $y = 5$ from 1st principle (3 marks)

2.5 a) $\int x^7 dx$ (1 mark) ✓

b) $\int \sqrt{x} dx$ (2 marks) ✓

c) Determine the value of $3x + 4 - \frac{5}{x}$ as x increases from -1 to 3. (3 marks)

2.6 Evaluate the determinant:

$$\begin{vmatrix} 1 & 3 & 2 \\ 4 & 5 & 7 \\ 2 & 4 & 8 \end{vmatrix}$$

(6 marks) ✓

2.7 Find the vector product of $\vec{P}$ and $\vec{Q}$ where $\vec{P} = 3\vec{i} - 4\vec{j} + 2\vec{k}$ and $\vec{Q} = 2\vec{i} + 5\vec{j} - 1\vec{k}$ (6 marks) ✓

$$\frac{4x^3 + 13x + 33}{2x + 3} = 2x^2 + 3x + 11 + \frac{66}{2x + 3}$$

$$\begin{pmatrix} 1 & 3 & 2 \\ 4 & 5 & 7 \\ 2 & 4 & 8 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 4 & 5 \\ 2 & 4 \end{pmatrix}$$

$$1(40 + 42 + 32) - (96 + 28 + 2)$$

$$114 - 144$$

SECTION D

Answer any FIVE (5) questions.

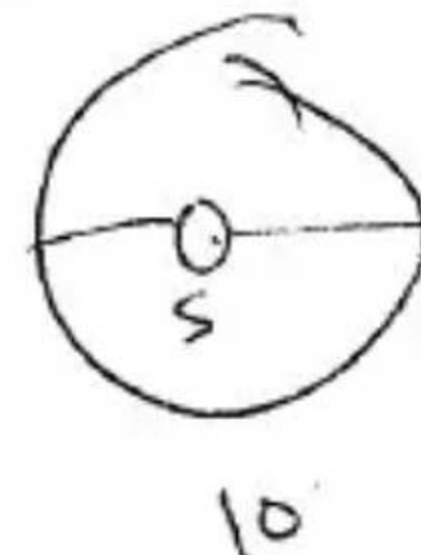
QUESTION 4

A vehicle has a wheelbase of 2.4m, a total mass of 2 tonne and the road wheels are 0.6m in diameter. When the rear axle is raised, the distance of the c.g. from the rear axle is 7.5m. The mass of the rear axle is 1.15 tonne and the mass on the front axle is 0.85 tonne. Determine the height of the c.g. above the ground level. (10 marks)

QUESTION 5

A hollow cast Iron Column of cross section 2.2m long has its internal sides measuring 150mm and is to carry a load of 1MN. If the safe working stress is to be limited to 120MN/m², Calculate:

- The area of the metal required the Load? (3 marks)
- The thickness of walls of column (4 marks)
- If the value of E of the material is 120GN/m², by how much will the column shorten under the load of 1MN. (3 marks)



QUESTION 6

A hydraulic press is operated by a force of 265N applied perpendicular to the end of a Lever. The lever ratio is 15 to 1. The pumping plunger is 25mm diameter and its stroke is 50mm. The ram is 200mm in diameter. Neglect friction, find:

- The load on the press (5 marks)
- The number of pumping strokes required to move the ram 50mm. (5 marks)

QUESTION 7

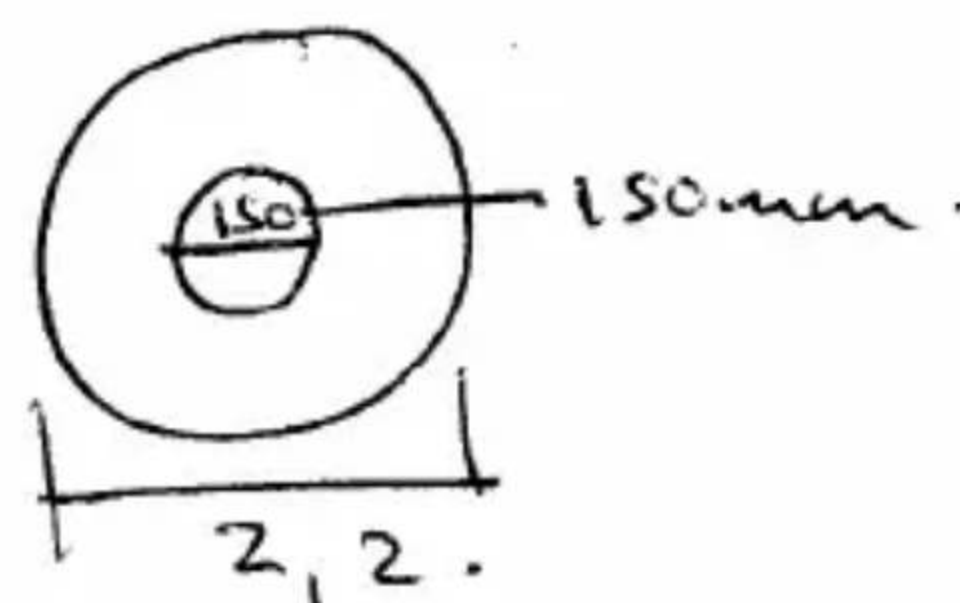
A steel component having a mass of 20kg and a temperature of 700°C is dropped into a tank containing 40kg of water at 15°C. Neglecting the heat energy absorbed by the tank and any loss of heat energy to the surrounding air, determine:

- The final temperature of water (6 marks)
- Heat energy absorbed by the water (4 marks)

Take specific heat capacity of steel and water as 0.48KJ/Kg k and 4.2KJ/Kgk respectively.

Stress

5



load = 1



ZIMBABWE

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR VEHICLE MECHANICS, AUTOMOBILE ELECTRICS AND
ELECTRONICS, DIESEL PLANT FITTING, MOTOR VEHICLE BODY
REPAIR, MOTOR CYCLE MECHANICS AND AUTOMOTIVE
PRECISION MACHINING**

**SUBJECT: Engineering Mathematics and
Science**

PAPER NO: 346/S03

DATE: 25 November 2009

TIME: 1400 – 1700 hours

REQUIREMENTS

A3 size drawing paper, drawing boards, pencils and drawing instruments.

INSTRUCTIONS TO CANDIDATE

1. This paper is divided into three sections. Each section carries 40 marks.
2. Answer one question from each Section.
3. Do not write your name on any answer script, except candidate and centre number.
4. Show all types of lines.
5. Use the shortest possible seams unless otherwise stated.
6. Candidates are allowed 10 minutes to read through the question paper before the examination starts.
7. Show projection symbol used.

This paper consists of 7 printed pages.

M.M

SECTION A

QUESTION 1

Find the value of $(7,9 \times 10^{-2}) - (5,4 \times 10^{-2})$. Express your answer in standard form. (1 mark)

QUESTION 2

Convert $111\ 0101_2$ to denary. (1 mark)

QUESTION 3

Find three consecutive numbers whose sum is 84. ✓ (1 mark)

QUESTION 4

Express 366° in radians. (1 mark)

QUESTION 5

Make b the subject of the formula $A = \frac{X-7}{B-b}$ (1 mark)

QUESTION 6

Find $\int 6\,du$ (1 mark)

QUESTION 7

Find the determinant of the following matrix.

$$\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} \quad (1 \text{ mark})$$

QUESTION 8

If vector $A = -2i + 4j$
 $B = i + 3j + 2k$

Find $\vec{A} + \vec{B}$ (1 mark)

QUESTION 9

Find the volume of a sphere whose diameter is 4mm. (1 mark)

QUESTION 10

Find the gradient of a curve $x^2 - 6 = y$ at a point $(2, -2)$ (1 mark)

SECTION B

QUESTION 1

Solve

a) $\log (x-1)+\log (x+1)=2 \log (x+2)$ (3 marks)

b) $\log 2^{(x+1)} = \log 3^{2x-5}$ (3 marks)

QUESTION 2

An engine has a 70mm bore and 82mm stroke and the connecting rod is 160mm long. Calculate;

a) The angle of obliquity of the connecting rod when the crank angle is 45° from t.d.c. (3 marks)

b) Piston travel when the crank angle is 45° from t.d.c. (3 marks)

QUESTION 3

Using matrices, solve the following simultaneous linear equation.

$$\begin{aligned} 3x + 2y &= 1 \\ x + 5y &= -4 \end{aligned} \quad (6 \text{ marks})$$

QUESTION 4

Differentiate from first principles.

a) $f(x) = 5x^3$ (3 marks)

b) $f(x) = 7x$ (3 marks)

QUESTION 5

Find the gradient of a curve $y = 7x^2 + 4x + 2$ at a point when $x = 2$.
Find also the equation of the normal to a tangent at that point. (6 marks)

QUESTION 6

Find the magnitudes and directions of the following vectors.

a) $3i + 4j$ (2 marks)

b) $-3i + 4j$ (2 marks)

c) $-i - 2j$ (2 marks)

QUESTION 7

Solve the following simultaneous equation.

$$\frac{3}{x} + \frac{2}{y} = 3$$

(6 marks)

$$\frac{5}{x} + \frac{3}{y} = -2$$

QUESTION 8

Find

a) $\int (3x^2 + 2x + 4) dx$

(2 marks)

b) $\int (8x^2 + 4) dx$

(2 marks)

c) $\int 3 dx$

(2 marks)

SECTION C

QUESTION 1

Define angular velocity.

(1 mark)

QUESTION 2

Explain what is meant by the resolution of a force into two components at right angles to each other.

(2 marks)

QUESTION 3

State one condition necessary for the equilibrium of a body acted on by a number of parallel forces.

(1 marks)

QUESTION 4

Find the amount of work done by a hydraulic hoist in lifting a vehicle of mass 1000kg through a distance of 3m.

(2 marks)

QUESTION 5

Complete the following: Power = Force x _____

(1 mark)

QUESTION 6

Name one vehicle component where allowance is made for volumetric expansion.

(1 mark)

QUESTION 7

What is meant by the 'coefficient of friction' between two surfaces. (1 mark)

QUESTION 8

Explain what is meant by the tensile strength of a material.

(1 mark)

SECTION D

QUESTION 1

A wheel initially at rest is subjected to a uniform angular acceleration of $2,5 \text{ rad/s}^2$ for 8s and is then immediately retarded uniformly until it comes to rest 50 seconds later. Calculate:

- the maximum angular velocity attained
- the total number of revolutions made by the wheel
- the angular retardation

(10 marks)

QUESTION 2

Calculate the amount of heat energy required to convert 10kg of ice at -20°C into steam at 100°C .

Specific latent heat of fusion of ice	$= 335\text{kJ/kg}$	
Specific latent heat of vapourisation of water	$= 2\,260\text{kJ/kg}$	
Specific latent capacity of ice	$= 2,1\text{kJ/kgK}$	(10 marks)

QUESTION 3

- State any three laws of dry friction.
- A multi-plate clutch has six annular frictional surfaces of 260mm external and 240mm internal diameter. The coefficient of friction is $0,3$. If the axial spring force on the pressure plate is 600N , calculate the power that can be transmitted at a speed of $2\,100\text{rpm}$.

(10 marks)

QUESTION 4

A screw jack has a double-start thread with a pitch of 3mm . If the Load to be raised is $1,2$ tonnes and the efficiency at this load is 20% , calculate

- Velocity ratio if the lever of the jack is 400mm .
- Mechanical Advantage $\frac{L}{E}$
- The torque required at the jack handle to raise the load.

(10 marks)

QUESTION 5

An engine has a cylinder bore of 64mm. The gudgeon pin has an outside diameter of 19mm and an inside diameter of 13mm, and the minimum cross-sectional area of the connecting rod is 320mm². If the maximum cylinder pressure at top dead centre is 3 300kPa, determine:

- the shear stress in the gudgeon pin.
- the maximum compressive stress in the rod. (10 marks)

QUESTION 6

A bus chassis, 5,4m long, consists of two side members and a number of cross-members. Each side member can be considered as a beam, simply supported at two points A and B, 3,6m apart, A being positioned 0,9m from the front end of the frame and subjected to the following concentrated loads:

Engine support (front) 2kN, Engine support (rear) 2,5kN, gearbox support 0,5kN and body WkN. The distance of these loads from the front end of the frame are respectively 0,6m, 1,8m, 2,4m and 3m. If the reaction at A is 8,5kN, determine:

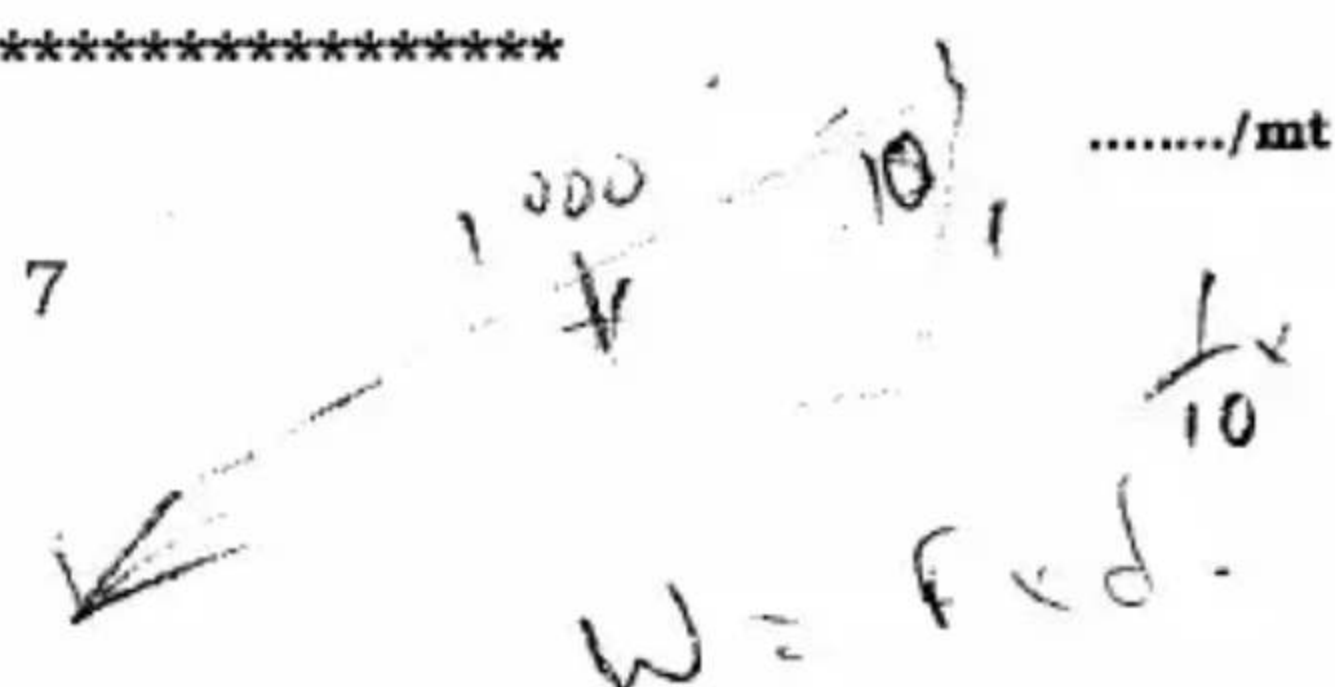
- the magnitude of the load W due to the vehicle body.
- the magnitude of the support reaction at B. (10 marks)

QUESTION 7

- A vehicle hauls a trailer at 80km/hr when exerting a steady pull of 500N at the tow-rope. Calculate the power required. (5 marks)
- The velocity of a body of mass 16kg is increased from 5m/s to 20m/s in 12s. Calculate the power required to produce the kinetic energy of the body. (5 marks)

*QUESTION 8

- Define work done. (2 marks)
- A motor car of mass 1 000kg is being propelled up a hill of gradient 1 in 10 at a steady speed of 64km/hr. Find the work done against gravity per minute. Neglect frictional resistance. (8 marks)



SECTION A

QUESTION 1

- 1.1 Determine the following product, giving the result in standard form.
 $(6.43 \times 10^3)(7.35 \times 10^4)$ (1 mark)
- 1.2 Factorise $x^2 + 3x - 10$ (1 mark)
- 1.3 Express the following in dinary form
 111.11_2 (1 mark)
- 1.4 In an electrical circuit the voltage V volts is given by $V = IR$. Find the voltage, when $I = 7.240$ amperes and $R = 12.57$ ohms, correct to 4 significant figures. (1 mark)
- 1.5 Solve the following equation for x
 $\lg x = 5$ (1 mark)
- 1.6 Integrate with respect to x
 $x^3 + 2x + 3$ (1 mark)
- 1.7 If $P = \begin{pmatrix} -1 & 2 \\ 3 & -4 \end{pmatrix}$ and $Q = \begin{pmatrix} 4 & 5 \\ 2 & 7 \end{pmatrix}$
Find $Q - P$ (1 mark)
- 1.8 Define a vector quantity. (1 mark)
- 1.9 Find the total surface area of a cone of radius 4.86cm and perpendicular height 10.58cm. (2 marks)

SECTION B**QUESTION 2**

- a) Solve the equation

$$\frac{4}{x-3} + \frac{2}{x} = \frac{6}{x-2}$$
(6 marks)
- b) The simple interest, \$I on a sum of money, \$P after T years is given by
 formula $I = \frac{PRT}{100}$
- i) Make T the subject of formula. (3 marks)
- ii) Find T if I = 51, P = 340 and R = 2 ½. (3 marks)
- c) i) If $y = 2x^2 - 5x + 3$,
 find $\frac{dy}{dx}$ and gradient at $x = 2$ (2 marks)
- ii) Find the equation of the tangent and normal at $y = 3x^3$
 at point (3, -3) (4 marks)
- d) i) $\int_0^1 (x^3 + x^{\frac{1}{2}} + 1) dx$ (3 marks)
- ii) Evaluate $[y^3]^{\frac{1}{2}}_{-\frac{3}{4}}$ (3 marks)
- e) A car travelling down the road increasing velocity from 8.9 m/s to 17.9 m/s in 20 sec.
 Calculate
- i) the distance travelled during this period. (3 marks)
- ii) the acceleration of the car. (3 marks)
- f) Use matrices to solve the following simultaneous equations

$$\begin{matrix} 2x + 5y = 1 \\ 3x - 2y = 30 \end{matrix}$$
(6 marks)
- g) Solve the indicial equation
 $x^{2.5} = 37.5$ correct to 4 significant figures. (6 marks)

SECTION C

QUESTION 3

- a) A wheel rotating at 5 rad/s accelerates uniformly at 0.5 rad/s² for 30 sec
Find the angular velocity after the 30 sec (2 marks)
- b) State Newton's second law of motion. (1 mark)
- c) A steel brake rod is 1.25m long and when subjected to a pull, the extension produced is 0.5mm. Calculate the strain in the rod. (1 mark)
- d) Define the term Power. (1 mark)
- e) A vehicle is towed along a straight level road by a force of 120N. Calculate the work done in towing the vehicle through a distance of 70m. (1 mark)
- f) Briefly explain the term equilibrant. (1 mark)
- g) Complete the following
 $\beta = 2x - - - - -$ (1 mark)
- h) A torque of 484Nm exists in the axle shaft of the driving wheels of a motor vehicle. The road wheels are 680mm in diameter, what is the tractive effort? (2 marks)

SECTION D

QUESTION 4

A uniform beam AB, 3m long, is simply supported at its ends. The force due to gravity on the beam is 2kN and loads of 3kN and 5kN are placed on the beam at distances 0.5m and 2m respectively from end A. Where must an additional load of 2kN be placed to make the support reactions equal. (10 marks)

QUESTION 5

A steel brake rod is 1.2m long and is subjected to a maximum load of 4.5kN. If the extension of the rod is not to exceed 0.384mm, determine

- a) The maximum stress produced in the rod. (4 marks)
- b) Suitable diameter for the rod. (6 marks)

QUESTION 6

A single plate clutch of 0.15m effective diameter is lined with material of coefficient of friction 0.35. If the total spring force is 2.5kN, calculate

- a) Maximum torque transmitted. (5 marks)
- b) The power transmitted at 3 000rpm. (5 marks)

QUESTION 7

A steel component having a mass of 20 kg and a temperature of 700°C is dropped into a tank containing 40kg of water at 15°C. Neglecting the heat energy absorbed by tank and any loss of heat energy to the surrounding air, determine

- a) The final temperature of water. (7 marks)
- b) The heat energy absorbed by the water. (3 marks)

Take the specific heat capacities of steel and water as 0.48kJ/kgK and 4.2kJ/kgK respectively.

QUESTION 8

A flywheel of diameter 360mm increases its speed uniformly from 630rev/min to 1050 rev/min in 11 sec

Calculate:

- a) The angular acceleration of the wheel, in rad/s². (4 marks)
- b) The number of revolutions made during the speed change. (4 marks)
- c) The linear acceleration of a point on the rim of the wheel in m/s². (2 marks)

QUESTION 9

- a) In a gearbox the constant – mesh gears had 14 and 30 teeth, respectively. The low gear on the main shaft had 28 teeth and the meshing layshaft gear had 16 teeth. The engine speed was 2 000 rev/min and the road wheel diameter was 0.56m. Calculate the road speed of the vehicle in this low gear. (6 marks)
- b) A gearbox provides a reduction of 4 to 1. What torque acts on the output shaft when the input torque is 70 Nm (Neglect frictional losses). (4 marks)

QUESTION 10

A screw jack has a 2 – start thread of 3mm pitch. An effort of 85N applied at a point 400mm along the handle is just sufficient to raise a mass of 800kg.

Calculate

- a) Torque on the handle in Nm. (3 marks)
- b) Velocity ratio of the jack. (3 marks)
- c) Mechanical advantage. (2 marks)
- d) Efficiency of the jack. (2 marks)

QUESTION 11

A single cylinder motor cycle engine has a connecting rod 120mm long and a stroke of 60mm.

Determine:

- a) Distance moved by piston from TDC when connecting rod reaches. Its maximum angularity relative to cylinder axis. (5 marks)
- b) The angle between the crank and connecting rod. (5 marks)

..... rg/wm



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SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR VEHICLE MECHANICS, AUTO-ELECTRICS, DPF, MVBR,
PRECISION MACHINISTS, MOTOR CYCLE MECHANICS**

**SUBJECT: Engineering Science and
Mathematics**

PAPER NO: 346/S03

DURATION: 3 Hours

OCTOBER/NOVEMBER 2015 EXAMINATION

REQUIREMENTS

Scientific Calculator

INSTRUCTIONS TO CANDIDATE

- 1. SECTION A: Answer ALL questions.**
- 2. SECTION B: Answer any FIVE (5) questions.**
- 3. SECTION C: Answer ALL questions.**
- 4. SECTION D: Answer any FIVE (5) questions.**

This paper consists of 5 printed pages.

SECTION A

Answer ALL questions.

QUESTION 1

- a) Factorise completely $20x^2 - 45$ (1 mark)
- b) Evaluate $(a^2 b^2 c)^{-2}$ (1 mark)
- c) Find the radius of a circle if its perimeter is 54cm (1 mark)
- d) Make E subject of the formula in $\frac{x+y}{E^2} = \frac{x^2+3x-4y}{x-y}$ (1 mark)
- e) Integrate with respect to x: $x^2 + 2x + 3$ (1 mark)
- f) Given that $\log 2 = 0.301$ and $\log 3 = 0.477$, find $\log 2^{7/9}$ (1 mark)
- g) Convert the following to base 2 346 (1 mark)
- h) Write the following number in preferred standard form 46 082 (1 mark)
- i) If matrix $A = \begin{bmatrix} -2 & -47 \\ 8 & -113 \end{bmatrix}$ Find $3A$ (1 mark)
- j) $A = -3i - 4j$
 $B = 8i + 12j$
 Find $B - A$ (1 mark)

SECTION B

Answer any FIVE (5) questions.

QUESTION 2

- a) Find the product of: $\begin{bmatrix} 4 & 7 & 6 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 5 \\ 9 \end{bmatrix}$ (6 marks)
- b) Nest the following polynomial $f(x) = 4x^4 - 6x^3 + 11x^2 + 7x + 2$
 $f(2) =$ (6 marks)
- c) Solve for $\log_4 = 1/256$ (6 marks)
- d) i) If $y = 2x^2 + 5$. Find the gradient when $x = 1.5$ (3 marks)
- ii) $\left(\frac{1}{\sqrt{x}}\right)^5$ (3 marks)

2 (

$$54 = 2\pi r^2$$

$$5(4x^2 - 9)$$

$$5(2x+3)(2x-3)$$

256

SECTION D**Answer any FIVE (5) questions.**QUESTION 4

A tensile force of 10KN is applied to a round steel bar with a diameter of 20mm and a length of 2m. Young's modulus of elasticity for this steel is 2000GPa. Calculate:

- a) Stress (5 marks)
b) The extension due to the tensile force (5 marks)

QUESTION 5

The distance S metres of a body travelling with constant acceleration a m/s² in a straight line is given by $S = ut + \frac{1}{2}at^2$ where u ms⁻¹ is the initial velocity and t seconds the time. Calculate the initial velocity and the acceleration given that $S = 36m$ at $t = 2s$ and $S = 124m$ when $t = 4s$ (10 marks)

QUESTION 6

- a) Calculate the heat energy required to melt completely 4kg of a copper alloy initially at a temperature of 80°C melting point of the alloy 1080°C specific heat capacity of the alloy = 0.4Kj/KgK specific latent heat of fusion of the alloy 210Kj/Kg. (6 marks)
- b) An exhaust from pipe is 1.5m long at 20°C. What will be its length when its temperature is raised to 370°C? The coefficient of linear expansion of the steel used is $12 \times 10^{-6}/K$ (4 marks)

QUESTION 7

Using a dynamometer having a torque arm length of 0.5m, the following brake loads were obtained during a Morse test on a test speed of 2520 rev/min:

Cylinder Firing	All Four	Nos 2;3;4	Nos 1;3;4	Nos 1;2;4	Nos 1;2;3
Brake Force (N)	180	118	125	127	121

Calculate:

- a) Brake Power (4 marks)
b) Indicated Power (6 marks)



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SCIENCE AND TECHNOLOGY DEVELOPMENT

HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)

NATIONAL CERTIFICATE

IN

MOTOR VEHICLE MECHANICS, AUTOMOBILE ELECTRICS AND
ELECTRONICS, MOTOR CYCLE MECHANICS, VEHICLE BODY
REPAIR DIESEL PLANT, FITTING AND PRECISION MACHINING

SUBJECT: Engineering Mathematics
and Science

PAPER NO: 346/SO3

DURATION: 3 Hours

MARCH/APRIL 2014 EXAMINATION

REQUIREMENTS

Scientific Calculator

INSTRUCTIONS TO CANDIDATE

- | | | |
|------------|--------------------------------|------------|
| SECTION A: | Answer ALL questions. | (10 marks) |
| SECTION B: | Answer any FIVE (5) questions. | (30 marks) |
| SECTION C: | Answer ALL questions. | (10 marks) |
| SECTION D: | Answer any FIVE (5) questions. | (50 marks) |

This paper consists of 5 printed pages.

M. M.

SECTION A

QUESTION 1

- 1.1 Convert 0.526_{10} to octal form. (1 mark)
- 1.2 Express $2\frac{1}{24}$ in decimal form to 2 decimal places (1 mark)
- 1.3 What percentage of 70 is 2? (1 mark)
- 1.4 $\log 2 + \log 6 - \log 4$ is what? (1 mark)
- 1.5 The volume of tank is 0.095m^3 . How many litres of petrol will be required to fill the tank? (1 mark)
- 1.6 The radius of a road curve is 150m. Find the angle turned through in 1 second in radians and degrees by a vehicle travelling at 72km/h. (2 marks)
- 1.7 Differentiate $y = x^5 - 2x^4 + 3x^3 - 4x^2 + 5$ (1 mark)
- 1.8 $\int (6x^5 + 4x^3 - 6)dx$ (1 mark)
- 1.9 $A = \begin{pmatrix} 4 & -3 \\ 1 & 2 \end{pmatrix}$ find the determinand (1 mark)

SECTION B

QUESTION 2

- 2.1 Solve the equations; $3x + 6y = 39$ and $4x - 3y = 8$ (6 marks)
- 2.2 Find the volume and capacity in litres of a tank with circular ends of diameter 0.3m and length 1.12m. (6 marks)
- 2.3 Divide $4x^3 + x^2 - 16x - 4$ by $x - 2$ (6 marks)
- 2.4 Differentiate with respect to x
 - a) $x^4 - 9x^3 + 6$ (2 marks)
 - b) $(x - 4)(x - 4)$ (2 marks)
 - c) $x^{-2}(1 + x)$ (2 marks)
- 2.5 Use matrices to solve the simultaneous equations

$$\begin{aligned} 4x - 3y &= 18 \\ x + 2y &= -1 \end{aligned}$$
 (6 marks)

2.6 If $P = i + 3j - k$ and $q = 2i + j + 4k$

Find :

- a) $p + q$ (2 marks)
- b) $p - q$ (2 marks)
- c) $|p + q|$ (2 marks)

2.7 a) Integrate $\int 7x^4 dx$ (2 marks)

- b) Find the value of c in (a) above when $x = 0$ and $y = 2$ and rewrite the integral replacing c with the calculated value (4 marks)

SECTION C

QUESTION 3

- 3.1 A vehicle is accelerated uniformly at 1.5m/s^2 from 27km/h to 81km/h . Calculate the time taken. (3 marks)
- 3.2 Find the work done when a force of 5970N lifts an engine through a vertical distance of 1.2m . (1 mark)
- 3.3 Given the frictional force $F = 3\text{N}$ and normal reaction $M = 10\text{N}$, find the coefficient of friction μ . (1 mark)
- 3.4 Define resultant force. (1 mark)
- 3.6 State the difference between indicated power and brake power. (1 mark)
- 3.7 Write down the formula for calculating Young's modulus of elasticity. (1 mark)
- 3.8 What is specific heat capacity? (1 mark)
- 3.9 Sketch a class 1 type of lever (1 mark)

SECTION D

QUESTION 4

- a) A simple screw jack has a thread of lead 5mm. If the effort applied at an effective radius of 210mm, determine the velocity ratio. (4 marks)
- b) It is found that the jack can lift a load of 500N which is W, when the effort P is 20N and that W is 1000N when P is 30N.

Presume that P and W are related by a law of the type $P = aW + b$, determine when the load is 1500N,

- i) The effort (2 marks)
- ii) The mechanical advantage (2 marks)
- iii) The efficiency of the jack (2 marks)

QUESTION 5

An engine has a stroke of 100mm and a connecting rod length of 200mm between centres. The crank is 45° past top dead centre when the force on the piston is 5000N. By means of graphical solutions, determine:

- a) The angle between the axis of the bore and the centre line of the con rod. (2 marks)
- b) The force acting along the connecting rod (3 marks)
- c) The force between the thrust force of the piston and the cylinder wall. (5 marks)

QUESTION 6

A car is travelling at a uniform speed of 50km/hr. It maintains this speed for 2 minutes. It is then uniformly accelerated to 100km/h in the next one minute and then immediately retarded and brought to rest after a total of 6 minutes travelling time. Find:

- a) The distance covered whilst accelerating (2 marks)
- b) The distance covered whilst braking (6 marks)
- c) The total distance covered (2 marks)

QUESTION 7

✓ A lorry hauls a trailer at 72km/h when exerting a steady pull of 800N on the tow rope. Calculate:

- a) i) The work done in 20 minutes
ii) in megajoules and in kilowatt hours (5 marks)

- ✓ b) The power required (5 marks)

QUESTION 8

A wheel 900mm diameter, whose axis is horizontal, rotates through an angle of 130° . What is the height of a point on its rim above what was originally the lowest position. Use trigonometry. (10 marks)

QUESTION 9

8000mm³ of methy, alcohol having a coefficient of volumetric expansion of $120 \times 10^{-5}/K$ is contained in a aluminium container that is just fills at $-10^\circ C$. Calculate the volume of alcohol that will overflow when its temperature rises to $40^\circ C$. Take coefficient of linear expansion of aluminium to be $24 \times 10^{-6}/K$. (10 marks)

✓ QUESTION 10

A horizontal beam AB is 6m long and is simply supported at A and B. Loads of 30kN and 60KN are carried at points 2m and 4m respectively from A.

- a) Find the magnitude of reaction of A and B. (4 marks)
b) If an extra load of 60KN is to be added to the beam in such a position that reactions are equal what will these reactions be at what distance from B must the 60KN force be situated? (6 marks)

✓ QUESTION 11

A wooden tray rest on a horizontal surface and it is found that with a mass of 3kg in the tray a horizontal force of 10N is just sufficient to cause the tray to slide. When the 3kg mass is replaced by a mass of 7kg it is found that the horizontal force required to cause the tray to slide is doubled.

Calculate:

- a) The coefficient of friction between the tray (with mass) and the surface. (6 marks)
b) The work done in moving the tray through a distance of 5m with an 11kg mass in the tray. (4 marks)

...../ab

$$60 \times x = 300$$

$$\frac{60x}{60} = \frac{300}{60}$$

$$x = 5$$



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NATIONAL CERTIFICATE

IN

**MOTOR MECHANICS, AUTO - ELECTRICS, DPF, PRECISION MACHINING,
MOTOR CYCLE MECHANICS, MVBR**

SUBJECT: Engineering Mathematics

PAPER NO: 346/13/S08

DURATION: 3 Hours

OCTOBER/NOVEMBER 2015 EXAMINATION

REQUIREMENTS

Scientific calculator

INSTRUCTIONS TO CANDIDATE

- 1. Section A: Answer all Questions**
- 2. Section B: Answer any five (5) questions**

This paper consists of 4 printed pages.

SECTION A

QUESTION 1

- a) Find the length of a straight line which passes through points:

A (2:-3)

B (5:4)

(2 marks)

- b) From the first principle solve

~~$(2x^2 + 4) \frac{\delta y}{\delta x}$~~

(6 marks)

- c) ~~$\int (3x^3 - 4x^2 + x) dx$~~

(3 marks)

QUESTION 2

- a) Solve the equation:

$$0.25 x^2 + 1.25x + 1.5 = 0$$

(3 marks)

- b) Simplify:

$$3\log + \log 4 + 2 \log 3 - \log 18$$

(3 marks)

- c) Find the inverse of matrix M:

$$\begin{pmatrix} 7 & 5 \\ 2 & 3 \end{pmatrix}$$

(3 marks)

QUESTION 3

- a) Rewrite in the nested form the following expression and then evaluate

$$f(x) = x^5 - 3x^4 + 4x^3 + 7x^2 + 2x + 8$$

$$\text{When } x = -1$$

(2 marks)

- b) i) Transpose for $x; y = ax^n$

(3 marks)

- ii) Find $(2x + 3)^2$

(2 marks)

- c) Write down the formula for calculating the area of a flat circular ring (annulus)

(2 marks)

- d) Evaluate, $\frac{2^2 \times 2^3}{24}$

(3 marks)

- e) A single cylinder engine has a connecting rod 120 mm long and stroke of 60 mm. Determine the distance moved by the piston from TDC when the

connecting rod reaches its maximum angularity relative to the cylinder axis (5 marks)

- f) Express 48_{10} to base 2 (binary) (3 marks)

SECTION B

QUESTION 4 ✓

- a) Factorise: $x^3 - 2x^2 - 5x + 6$ (6 marks)

- b) Solve by completing the square the following equation

$$x^2 + 3x - 4 = 0 \quad (6 \text{ marks})$$

QUESTION 5

- a) A petrol tank 0.5m long has spherical ends of radius 0.3m. Determine:

- i) The capacity of the tank. (6 marks)
- ii) The mass of petrol when the tank is full (4 marks)
- iii) The cost of filling the tank if a litre costs \$1.38. (2 marks)

QUESTION 6 ✓

- a) Determine the constant of integration:

$$\int (4x - 5 + \frac{6}{x^2}) dx \text{ when } x = -\frac{1}{2}, y = 5 \quad (6 \text{ marks})$$

- b) ✗ Evaluate $\int_{-1}^3 (3x^2 - 4x + 5)dx$ (6 marks)

QUESTION 7

- a) Use matrices to solve the simultaneous equations:

$$\begin{aligned} 4a + 5b + c &= 2 \\ a - 2b - 3c &= 7 \\ 3a - b - 2c &= 1 \end{aligned}$$

(12 marks)

QUESTION 8 ✓

- a) Find the coordinates of the points on the following curve where the gradient is zero

$$y = (2+x)(3-4x) \quad (6 \text{ marks})$$

- b) Find the equation of the tangent to $y = x^2$ at the point (1:1) and the equation of the tangent to $y = \frac{1x^3}{6}$ at the point $(2; \frac{4}{3})$. Show that the tangents are parallel. (6 marks)

QUESTION 9

- a) Solve the indicial equation

$$7(14.3^{x+5}) \times 6.4^x = 294$$

Correct of 4 significant figures

(6 marks)

- b) Prove without using logs

$$\frac{\sqrt{27} + \log \sqrt{8} - \log \sqrt{125}}{\log 6 - \log 5} = \frac{3}{2}$$

(6 marks)

QUESTION 10

- a) A certain car has a wheel base of 4m and wheel track of 1.6m. When the car is turning on a certain radius the steering angle of the inner wheel is 22°

Calculate

- The angle of the outer wheel
 - The length of the radius (measured to the centre of the rear axle)
- (6 marks)

- b) The starting point of a hill climb course has an angle of depression of 12° when viewed from the finishing line which is 100 m above the level of starting point

- Find the distance of the course assuming that it is 12% longer than the gradient. (6 marks)

...../dn



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR VEHICLE MECHANICS, MOTOR CYCLE MECHANICS MVBR,
PRECISION MACHINING, AUTO ELECTRICS, DIESEL PLANT FITTING**

**SUBJECT: Automotive Engineering
Mathematics**

PAPER NO: 346/13/S08

DURATION: 3 Hours

OCTOBER/NOVEMBER 2014 EXAMINATION

REQUIREMENTS

Calculator

INSTRUCTIONS TO CANDIDATE

**SECTION A: Answer ALL questions. Each question carries
one mark.**

**SECTION B: Answer ALL questions. Each question carries
5 marks.**

**SECTION C: CHOOSE any FIVE (5) questions. Each question
carries 10 marks.**

Show all the working.

This paper consists of 4 printed pages.

PAPER NO: 346/13/SO8 – AUTOMOTIVE ENGINEERING
MATHEMATICS

- f) Find the value of $\frac{3^3 \times 4^4}{2^3 \times 9^2 + 4^2 \times 27}$ (5 marks)
- g) Solve $4x - 18 = 3y$
 $1 + x + 2y = 0$ (5 marks)
- h) In a triangle XYZ, $\angle Y$ is a right angle, the hypotenuse is 13cm long and the side XY is 5cm long. Find the length of the side YZ. (5 marks)

SECTION C

Choose any FIVE (5) questions from this Section. Each question carries ten (10) marks.

QUESTION 3

- a) The distance S metres moved by a body in t seconds is given by
 $S = 3t^3 - \frac{11}{2}t^2 + 2t + 5$
- Find:
- i) Its velocity after t seconds
 - ii) Its velocity at the start and after 4 seconds
 - iii) The value of t when the body comes to rest
 - iv) Its acceleration after t seconds
 - v) Its acceleration after 2 seconds
 - vi) The value of t when the acceleration is 16m/s^2 (10 marks)
- b) i) Evaluate $\frac{27^{0.5} \times 243^{0.5}}{243^{0.8}}$
- ii) Evaluate $\frac{D}{2} = \sqrt{R^2 + \frac{(R^2 + r^2 h^2)^2}{2h}}$ when $R = 75$, $r = 51$ and $h = 28$ (10 marks)
- c) Solve for x $2^{2x+8} - 32(2^x) + 1 = 0$ (10 marks)
- d) Differentiate from 1st principle $y = 6x^2 + 3x - 4$ (10 marks)
- e) Solve for x in the general form of a quadratic equation
 $ax^2 + bx + c = 0$ (10 marks)
- f) ✕ Integrate the following functions
- i) $\int 2 + 3x - 4x^2 dx$
 - ii) $\int \frac{1}{2}x - \frac{2}{\sqrt{x}} - 1 dx$
 - iii) $\int_1^3 (4x - 3)^2 dx$; (10 marks)

PAPER NO: 346/13/SO8 – AUTOMOTIVE ENGINEERING
MATHEMATICS

g) i) Determine the single matrix for A.B where

$$A = \begin{pmatrix} 2 & 4 \\ 1 & -3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & -7 \\ 4 & -5 \end{pmatrix}$$

ii) Solve the equation $\frac{y+2}{4} + \frac{3}{y-1} = 5$ by completing the square
give the answer to correct 4 significant figures. (10 marks)

...../ab



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MINISTRY OF HIGHER AND TERTIARY EDUCATION

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR VEHICLE MECHANICS / AUTOMOBILE ELECTRICS AND
ELECTRONICS / MOTOR CYCLE MECHANICS / MOTOR VEHICLE
BODY REPAIR / DIESEL PLANT FITTING AND PRECISION
MACHINING**

**SUBJECT: Engineering Mathematics and
Science**

PAPER NO: 346/S03

OCTOBER / NOVEMBER 2013 EXAMINATION

REQUIREMENTS

Scientific calculator, graph paper, pen, pencil and answer booklet.

INSTRUCTIONS TO CANDIDATE

1. Section A: Answer all questions (10 marks).
2. Section B: Answer any five (5) questions (30 marks).
3. Section C: Answer all questions (10 marks).
4. Section D: Answer any five (5) questions (50 marks).

This paper consists of 7 printed pages.

MM

5

SECTION B

QUESTION 2

- a) Find the remainder in the polynomial

$$6x^3 + 7x^2 + 2x + 2 \div 2x - 1$$

(6 marks)

- b) Evaluate

$$\frac{\log_x 32 - \log_x 4 + \log_x 8}{\log_x 256}$$

(6 marks)

- c) An engine has a stroke and con rod length of 172cm as measured from the centres of the bearings. Determine the distance moved by the piston from t.d.c. and the angle of obliquity when crank is at 65° from t.d.c. (6 marks)

- d) Find the differential coefficient of $f(x) = 3x^3$ from first principles. (6 marks)

- e) Determine the value of $3x + 4 - \frac{5}{x}$ as x increases from -1 to 3 . (6 marks)

- f) Use matrices to solve the simultaneous equations (The inverse method)

$$4a - 3b = 18$$

$$a + 2b = -1$$

(6 marks)

- g) $a = 3i - 4j + 5k$

$$b = i - 5j + 7k$$

Obtain

i) $a + b$

(2 marks)

ii) $a - b$

(2 marks)

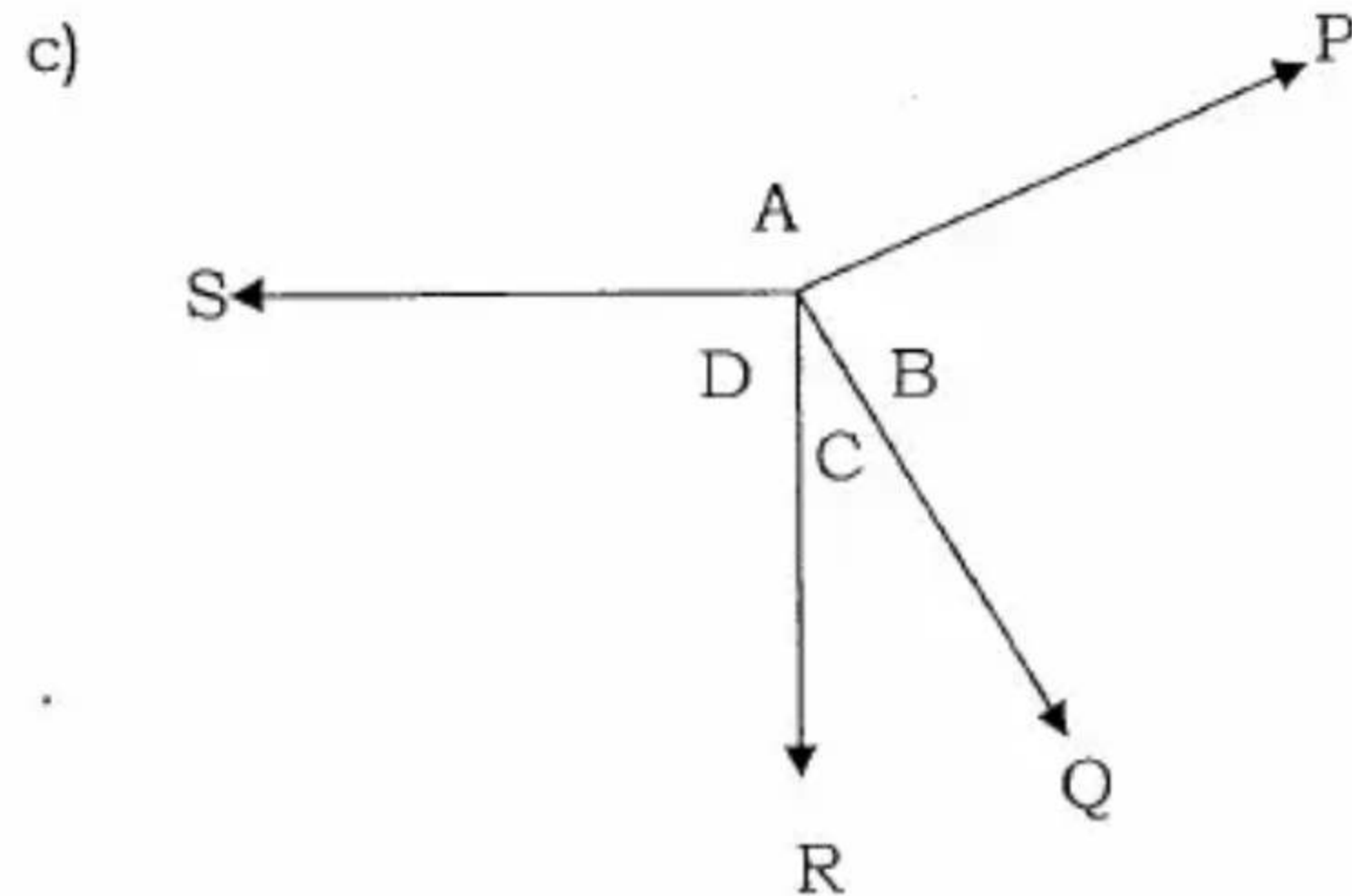
iii) $|a - b|$

(2 marks)

SECTION C

QUESTION 3

- a) Find the length of an arc of a circle of radius 4.23cm when the angle subtended at the centre is 1.46 radians. (1 mark)
- b) A load of 200kg is suspended stationary on the end of a wire rope. Determine the tension in the rope. (2 marks)



The fig above shows the space diagram of four forces acting at a point, the forces are not in equilibrium. Construct a sketch of a polygon of forces and draw in vector representing the force required to maintain equilibrium. (1 mark)

- d) Find the workdone when a force of 5970N is required to lift an engine through a vertical distance of 1.35m. (1 mark)
- e) A vehicle hauls a trailer at 72km/h when exerting a steady pull of 800N at a low rope. Calculate the power required. (1 mark)
- f) Express the heat required to raise the temperature of 5kg of water from 4°C to 80°C in kJ. (1 mark)
- g) It is found that a vehicle having a mass of 900kg can be moved slowly and steadily along a horizontal surface by a pull of 5.74kN with the wheels locked. Calculate the coefficient of friction between tyres and road. (1 mark)
- h) Convert $162 \text{GN/m}^2 = \text{___MPa}$ (1 mark)
- i) A motor vehicle of mass 1650kg is travelling on the level at a speed of 72km/h. Determine its kinetic energy. (1 mark)

SECTION D

QUESTION 4

A motor vehicle of total mass $2Mg$ is uniformly accelerated from rest to a speed of $54km/h$ in a distance of $150m$ up an incline of 1 in 15 . Each axle of the vehicle together with the wheels has a mass of $150kg$ and a radius of gyration of $0.38m$. the wheel tread diameter is $0.914m$ and the resistance to motion is $400N$. Calculate the tractive effort required.

(10 marks)

QUESTION 5

A screw lock has a thread of $6mm$ pitch. An effort of $80N$ applied at a radius of $420mm$ is just sufficient to move a load of $8.8kN$.

Calculate

- the efficiency of the Jack. (4 marks)
- the work done on the load to raise it by $50mm$. (3 marks)
- the work done in overcoming friction when raising the load by $50mm$. (3 marks)

QUESTION 6

A morse test on a 5 cylinder petrol engine gave the following results when the speed was maintained at $2\,000\text{ rev/min}$.

Cylinder not firing	None	No. 1	No. 2	No. 3	No. 4	No. 5
PB (kW)	40.65	32,44	31,32	32,8	32,1	31,7

From the above results, calculate

- indicated power
- friction and pumping losses
- the mechanical efficiency of the engine

(10 marks)

QUESTION 7

At a certain instant, a horizontal rope towing a vehicle makes an angle of 60° with the front axle. If the force on the tow rope is 1300N. Calculate

- a) the effective force pulling the vehicle forward.
- b) the force pulling the vehicles side ways (10 marks)

QUESTION 8

Determine the amount of heat energy to be removed from 2kg of water at 40°C to produce ice at a temperature of -5°C .

Specific latent heat fusion of ice = 335kJ/kg

Specific heat capacity of ice = 2.1kJ/kgk

Specific heat capacity – water = 4.2kJ/kgk (10 marks)

QUESTION 9

A Journal bearing is subjected to a load of 90N. The diameter of the journal is 70mm and the coefficient of friction between the journal and bearing is 0.15. Given that the engine is running at 1500 revs/min determine:

- a) friction force between Journal and bearing
- b) frictional torque
- c) heat generated per minute
- d) power lost to friction (10 marks)

QUESTION 10

In a laboratory experiment a wire of cross-sectional area 0.5mm^2 and original length 2m was subjected to increasing loads. The extension was measured for each load with the following results:

Load N	0	50	100	150	200	250	300
Extension	0	1	2	3	4	5	6

Plot graphs of load against extension and stress against strain and state what may be deduced from their character. From one of these graphs, determine the value of Young's modulus of elasticity for the material of the wire. (10 marks)

QUESTION 11

A vehicle has a wheel base of 4m and the distribution of load when the vehicle is at rest is as follows:-

Front axle 4500N

Rear axle 6000N

Determine how far from the center of gravity lies from the front axle. (10 marks)

...../pm



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MINISTRY OF HIGHER AND TERTIARY EDUCATION

**HIGHER EDUCATION EXAMINATIONS COUNCIL
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IN

**MOTOR VEHICLE MECHANICS, AUTOMOBILE ELECTRICS AND
ELECTRONICS, DIESEL PLANT FITTING, MOTOR VEHICLE BODY REPAIR,
MOTOR CYCLE MECHANICS AND PRECISION MACHINING**

**SUBJECT: Engineering Science
And Mathematics**

PAPER NO: 346/S03

NOVEMBER/DECEMBER 2010 EXAMINATION

REQUIREMENTS

Pen, Pencil, Calculator, Drawing instruments

INSTRUCTIONS TO CANDIDATE

- SECTION A:** Answer ALL questions.
- SECTION B:** Answer any five (5) questions.
- SECTION C:** Answer ALL questions.
- SECTION D:** Answer any five (5) questions.

This paper consists of 5 printed pages.

SECTION A

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 1 MARK.

QUESTION 1

- 1.1 Determine the denary equivalent of 476.352_8 .
 1.2 Transpose and make 't' the subject, $s = \frac{1}{2}(u + v)t$ ✓
 1.3 If $f(x) = x^3 + 3x^2 - 13x - 10$, find $f(3)$. ✓
 1.4 If $\log 3$ is 0.47712, find $\log 9$ without using a calculator.
 1.5 What is the relative density of mercury given that its density is 13600 kg/m^3 .
 1.6 Given that $\sin \theta = \frac{\sqrt{3}}{2}$ and $\cos \theta = \frac{1}{2}$ what is $\tan \theta$ and the angle θ .
 1.7 $f(x) = 2x^3 + 1$. Find its derivative with respect to x. ✓
 1.8 Find $\int 3x^3 dx$ ✓
 1.9 Evaluate

$$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \checkmark$$

- 1.10 Find the sum of the following vectors $\vec{AB} + \vec{BC} + \vec{CA} = \dots\dots\dots$

SECTION B

ANSWER ANY FIVE (5) OUT OF SEVEN (7) QUESTIONS. EACH QUESTION CARRIES 6 MARKS.

QUESTION 2

- 2.1 a) Determine the remainder that would occur if
 $(4x^3 - 5x^2 + 7x - 3)$ when divided by $(x - 3)$ (3 marks)
 b) Rewrite in nested form and determine the value of the function for the value of x stated.

$$f(x) = 3x^3 - 2x^2 - 5x + 3 \quad [x = 4] \quad (3 \text{ marks})$$

- 2.2 a) Solve the following equation

$$8 \times 15.4^{2x+1} = 5.9^{4x} \quad (3 \text{ marks})$$

- b) Express the following without logarithms.

$$\log R = 1 + \frac{1}{3} \log M + 3 \log N \quad (3 \text{ marks})$$

- 2.3 If $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 1 \end{bmatrix}$ Find

$\therefore \log 8 \times \log 15.4 = 4 \times \log 5.9$
 $2 = 1 + m \frac{1}{3} + n^3$
 $1 + 1 \log m \frac{1}{3} + \log n^3$
 $1 + \sqrt[3]{n^3}$
 $\log(m \frac{1}{3} n^3) = 1$

- a) $A.B$ (2 marks)
- b) $A.B^T$ (3 marks)
- c) $|A|$ (1 mark)
- 2.4 A taper of 2 degrees included angles is required on a shaft, the diameter of the small end being 13.75mm. Calculate the length of the shaft necessary to give a large diameter of 14.37mm. (6 marks)

2.5 If $\vec{a} = 5i + 22j + 5k$
 $\vec{b} = i + 2j + 3k$
 $\vec{c} = 4i + 3j + 2k$

Find

- a) $\vec{a} + \vec{b}$ (2 marks)
- b) $\vec{a} - \vec{c}$ (2 marks)
- c) $\vec{b} + \vec{c}$ (2 marks)
- 2.6 a) State the equation of a straight line and differentiate with respect to x . (3 marks)
- b) Find the derivative of the polynomial $3x^4 + 2x^3 - x^2 + 1$. (3 marks)
- 2.7 a) Find $\int (x^2 - 2x) dx$ (2 marks)
- b) If an equation has a gradient function $(x + 4)$ and passes through the point $(2,3)$ find the equation of the curve. (4 marks)

SECTION C

ANSWER ALL QUESTIONS. EACH QUESTION CARRIES 1 MARK.

QUESTION 3

- 3.1 Given that linear velocity of a point on the edge of a flywheel rotating at 66 rad/s is 33m/s. Calculate the radius of the flywheel. (1 mark)
- 3.2 Define concurrent forces. (1 mark)
- 3.3 State in terms of x and y , the components of force F , inclined at θ to the horizontal. (1 mark)



- 3.4 Calculate work done in moving a body for 7m against resistance of 10N. (1 mark)
- 3.5 State the principle of moments. (1 mark)
- 3.6 State one method of heat transfer. (1 mark)
- 3.7 What is the power in doing work of 60KJ on one (1) minute? (1 mark)
- 3.8 Calculate the coefficient of friction required when a block of mass 20kg is hauled by a 40N horizontal force. (1 mark)
- 3.9 A bolt of 30mm² cross-sectional area is subjected to double shear by a force of 6KN. Calculate the shear stress. (1 mark)
- 3.10 A piston is moved at a uniform velocity of 7,5m/s against a resistance of 200N. Find the power developed. (1 mark)

SECTION D

ANSWER ANY FIVE (5) OUT OF EIGHT (8) QUESTIONS. EACH QUESTION CARRIES 10 MARKS.

QUESTION 4

A bus having a mass of 3Mg is travelling along a level road at 60km/h. The resistance to motion is 10N per kilo Newton of weight. Calculate

- a) the tractive effort required (4 marks)
- b) the power and (3 marks)
- c) the work done in 40 minutes in megajoules. (3 marks)

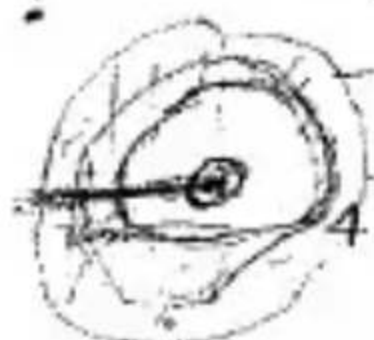
QUESTION 5

A lorry of mass 10Mg reaches a speed of 63km/h in 20s, starting from rest with uniform acceleration. It travels 2km at this speed and is then brought to rest in a distance of 125m. Determine:

- a) the momentum at 63km/h. (2 marks)
- b) the accelerating and braking forces required. (8 marks)

QUESTION 6

A single plate clutch transmits a torque of 240Nm. The total spring force acting on the pressure plate is 3360N and the coefficient of friction of the lining material is 0,3. Calculate the outside diameter of the lining if it had a width of 26mm. (10 marks)



Handwritten calculations and diagrams for Question 6:

Diagram: A horizontal line representing the clutch disc with a central hub. A force of 40N is applied to the right at the end of the disc. The mass is 20kg.

Equations:

$$\frac{1}{2}(r_1 + r_2) = 0,11$$

$$r_2 = r_1 + 26$$

$$r_1 + r_1 + 26 = 0,238$$

$$2r_1 + 26 = 0,238$$

$$2r_1 = 0,238 - 26$$

$$r_1 = \frac{0,238 - 26}{2}$$

- 3.4 Calculate work done in moving a body for 7m against resistance of 10N. (1 mark)
- 3.5 State the principle of moments. (1 mark)
- 3.6 State one method of heat transfer. (1 mark)
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SECTION D

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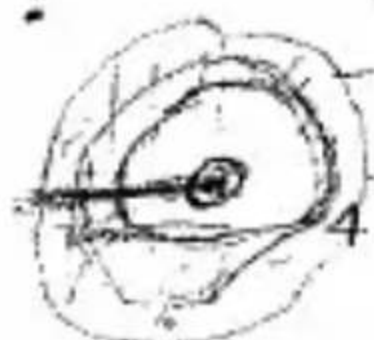
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A single plate clutch transmits a torque of 240Nm. The total spring force acting on the pressure plate is 3360N and the coefficient of friction of the lining material is 0,3. Calculate the outside diameter of the lining if it had a width of 26mm. (10 marks)



Handwritten calculations and diagrams for Question 6:

Diagram: A horizontal line representing the friction lining with a force $F = 3360\text{N}$ acting downwards at the center. A torque $T = 240\text{Nm}$ is indicated at the right end.

Equations:

$$\frac{1}{2}(r_1 + r_2) = 0,11$$

$$r_2 = r_1 + 26$$

$$r_1 + r_1 + 26 = 0,238\text{m}$$

$$2r_1 + 26 = 0,238\text{m}$$

$$2r_1 = 0,238\text{m} - 26$$

$$r_1 = \frac{0,238\text{m} - 26}{2}$$

Preface to 1993 edition

The New General Mathematics series has been revised to reflect the content and philosophy of mathematical education in Zimbabwe's secondary schools. Books 1 and 2 provide a full course at Junior Certificate Level. Books 3 and 4 contain a substantial course leading to the General Certificate in 'O' Level Mathematics. In Books 3 and 4, special emphasis is given to the development of mathematical skills which school leavers need to acquire for their everyday lives, and for further study and training. To achieve this there are extensive new sections on consumer arithmetic and on the use of calculators.

In order to develop problem solving skills and provide opportunities for independent thought, users of Books 3 and 4 will find sections which include 'non-routine problems'. Students and their teachers are urged to approach these with an open mind, and with the aim of finding unexpected joy and self-fulfilment in mathematics.

The need to consolidate and practice problem solving skills is maintained through periodic revision tests. These can be used to diagnose performance in mathematics and provide a basis for further improvement.

Whilst revising Books 3 and 4, opportunities were used to make corrections and to update statistical information. The authors and publishers are grateful to the Central Statistical Office, Harare, for providing valuable data. Above all, we are grateful to the many readers who have made helpful suggestions and who have provided so much encouragement.

M.F. Mawhood

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Acknowledgements

First published 1985
Second edition 1993
Nineteenth Impression 2011

ISBN 0 582 08963 8 (without answers)
ISBN 0 582 08964 6

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Printed by CTP Printers, Cape Town

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Associated companies, branches and representatives throughout the world

Longman House,
Burnt Mill, Harlow, Essex CM 20 2JE, UK.

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M.F. Macrae

1992

Longman Group UK Limited,
Longman House,
Burnt Mill, Harlow, Essex CM 20 2JE, UK.

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HARARE POLYTECHNIC

AUTOMOTIVE ENGINEERING DIVISION

2016 END OF TERM 1 TESTS- ENGINEERING MATHS & SCIENCE

INSTRUCTIONS

Section A: answer all questions

Section B: answer 5 questions

Section C: answer all questions

Section D: choose any 5 questions

SECTION A

QUESTION 1

- a. Transpose for F in $\frac{1}{F} + \frac{1}{v} = \frac{1}{L}$ **[1]**
- b. Define irrational number **[1]**
- c. Convert 67029 to preferred standard form **[1]**
- d. Write 64 in index for m using base 4 **[1]**
- e. Factorise $(x - 3)^2$ **[1]**
- f. Evaluate $2 \times 3 - 5 + 7 \div 3$ **[1]**
- g. Convert 17 to duodecimal **[1]**
- h. Expand $(2x + 1)(x + 3)$ **[1]**
- i. Solve for x in $x^2 - 9x + 18 = 0$ **[1]**
- j. Find $f(-2)$ if $f(x) = x^3 - x + 1$ **[1]**

SECTION B- ANSWER ANY 5 QUESTIONS

Question 2

- a.**
- i.** If $f(x) = x^4 - x^2 + 2x - 5$ find $f(-2)$ **[1]**
 - ii.** Express $2x^4 + 3^2 + 4$ in nested form **[2]**
 - iii.** Prove that $4x^4 + 19x^3 + 24x^2 + x - 10 \div x - 1 = 4x^3 - 15x^2 + 9x + 10$ by long division **[3]**
- b.** Convert
- i.** 11.101_2 to base 16 straight **[2]**
 - ii.** 11001_{12} to hexadecimal **[4]**
- c.** The period of a compound pendulum is given by $2\pi \sqrt{\frac{h^2 + k^2}{gh}}$ make k the subject of the formula taking g as 10 **[6]**
- d.** Use Pascal's Triangle to expand $(2x-4)^4$ **[6]**
- e.**
- i.** Use factor theorem to solve $x^4 + x^3 - 9x^2 + x + 10 = 0$ **[4]**
 - ii.** Express in nested form and evaluate for $f(-2)$ in $2x^4 - 3x^2 + 2$ **[2]**
- f.** Solve simultaneously
- i.** $3x - 2y - 13 = 0$
 - ii.** $2x + 5y + 4 = 0$
 - iii.** $\frac{x}{2} + \frac{y}{3} - 4 = 0$
 - iv.** $\frac{x}{6} - \frac{y}{9} = 0$ **[3]**
- g.** Solve by completing the square $15x^2 + 2x - 8 = 0$ **[6]**

SECTION C- ANSWER ALL QUESTIONS

QUESTIONS 3

- a.** State the principle of conservation of energy **[1]**
- b.** State the principle of moments **[1]**
- c.** Give 2 conditions for a body under a number of parallel forces to be in equilibrium **[2]**
- d.** What is the momentum of a 60kg body moving at 16m/s **[1]**
- e.** What is the length of an arc of a circle of radius 21cm subtended at an angle of 30° **[2]**
- f.** Define concurrent forces **[1]**
- g.** What is the horizontal component of a force of 70N inclined at 60° from the vertical **[1]**
- h.** Give one factor affecting braking efficiency **[1]**

SECTION D – CHOOSE ANY 5 QUESTIONS

QUESTION 4

A bus chassis is 6m long consists of two side members and a number of cross members. Each side member considered as a beam simply supported at two points **A** and **B** 3m apart. **A** beam is 1m from the front end of the frame and subject to the following loads engine support (front) 2kN, support (rear) 2,5kN, gearbox support 0,5kN and body WkN. The distance of loads from rear are 0,6m, 2m, 2.4 and 3m respectively. If the reaction at **A** is 8,5kN.

- i. Find magnitude of load W due to the vehicle body **[6]**
- iii. Magnitude of support of reaction **B**. **[4]**
show diagram

QUESTION 5

- a. A wheel rotating at 5 rads/s accelerates uniformly at 0.5 rad/s^2 for 30 seconds. Find :-
 - i. Angular velocity after 30 seconds **[2]**
 - ii. Angle turned through by the wheel during this time **[2]**
- b. If the wheel is brought to rest from this speed in 15 sec calculate:-
 - i. Braking force **[3]**
 - ii. Braking efficiency of the brakes **[3]**

QUESTION 6

Calculate the magnitude of resultant R and its direction by analytical method. **[10]**

QUESTION 7

A vehicle exerting a downward force of 19930N descends an incline of 1 in 15 at uniform speed with the engine switched off

- a. What is the total resistance to its motion **[5]**
- b. What power would be required to drive the vehicle at 63km/hr against this resistance on a horizontal road **[5]**

QUESTION 8

A car starting from rest and moving with uniform acceleration covers 50metres from the start in 5 seconds

- i. What is the velocity at this instant **[4]**
- ii. If the car now continues moving with uniform speed for 2 minutes and then brakes are applied so that it comes to rest with uniform retardation in 3 seconds . Find total distance covered **[6]**

QUESTION 9

A vehicle has a wheel base of 4m and the distribution of load when the vehicle is at rest is as follows:-

Front scale $4\,500\text{N}$

Rear axle $6\,000\text{N}$

Determine how far from the front axle lies the centre of gravity **[10]**

QUESTION 10

A bus having a mass of 3Mg is travelling along a level road at 60km/hr . The resistance to motion is 10N per kilo Newton of weight. Calculate

- a. Tractive effort required **[3]**
- b. Power **[2]**
- c. Work done in 40 minutes **[3]**
- d. Kinetic energy of the bus **[2]**

HARARE POLYTECHNIC

AUTOMOTIVE ENGINEERING DIVISION

2016 END OF TERM 1 TESTS- ENGINEERING MATHS

INSTRUCTIONS

Section A: answer all questions

Section B: answer all questions

SECTION A

QUESTION 1

- a. Make t the subject of the formula $s = \frac{1}{2}(u + v)t$ **[1]**
- b. Convert the denary decimal to octal 0.526 **[1]**
- c. How many litres are there in 0.095m^3 **[1]**
- d. Solve for x and y in
- i. $3x + 6y = 39$
- ii. $4x - 3y = 8$ **[4]**
- e. Calculate the volume of a cylinder with height of 1,12m and diameter of 0.30m **[2]**
- f. Add 66_8 **[1]**
- $$\begin{array}{r} 66_8 \\ +32_8 \\ \hline \end{array}$$
- g. Factorise $2x^2 - 7x + 3 = 0$ **[2]**
- h. $f(x) = 3x^3 - 2x^2 - 5x + 3$ ($x = 4$) **[5]**
- i. prove that $(x - 3)$ is a factor of $2x^2 - 5x + 1$ **[3]**
- j. $(x - 2)(x + 2)$ **[2]**

k. write 0.0007845 to 3 sf [1]

l. $(1763 \times 10^5) + (20 \times 10^5)$ [1]

m. $\frac{1}{x-1} + \frac{1}{x+3} + \frac{1}{x+2}$ [4]

n. $x^2 - 9x + 18 = 0$ solve for x [4]

o. convert 11,11001 to base 16 [3]

p. $x^3 - x^2 + ax + b$ solve for a and b [5].

If f(x) is divided by $(x - 1)$ the remainder is -8 and if f(x) is divided by $(x + 1)$ the remainder is 0

SECTION B- ANSWER ALL QUESTIONS

2. Solve for x

$$ax^2 + bx + c = 0 \text{ [10]}$$

3a. $4x^3 + 13x + 33 \div (2x + 3)$ [5]

b. find x

$$\frac{1}{4}x^2 + 1\frac{1}{4}x + 1\frac{1}{2} = 0 \text{ [5]}$$

4. Change 2E 3, 4D₁₆ to denary via binary and octal [10]

5. The father is 4 times older than his son. Five years back the product of their years was 234. Find their present age [10]

6. Factorise completely

$$2x^4 - 5x^3 - 15x^2 + 10x + 8 = 0 \text{ [10]}$$

7. $\frac{3}{x} + \frac{2}{y}$

$$\frac{5}{x} + \frac{3}{y} = -2 \text{ find x and y [10]}$$

HARARE POLYTECHNIC

AUTOMOTIVE ENGINEERING DIVISION

ENGINEERING SCIENCE- END OF TERM 1 TEST

INSTRUCTIONS

Section A: Answer all questions

Section B; answer ANY 5 questions

SECTION A

QUESTION 1

- a. Define the following terms
 - i. radian
 - ii. angular displacement
 - iii. angular velocity
 - iv. angular acceleration [4]**
- b. Derive the relationship between linear and angular velocity **[5]**
- c. Write the corresponding equations relating to angular motion of the equations below
 - i. $s = ut + \frac{1at^2}{2}$
 - ii. $v^2 + u^2 + 2as$ **[2]**
- d. State Newton's three laws of motion **[6]**
- e. Give the SI unit for mass, weight and momentum **[3]**
- f. Define the following
 - i. Kinetic energy
 - ii. Potential energy
 - iii. Power
 - iv. Work [4]**
- g. Differentiate between equilibrant and resultant force **[4]**

- h. Sketch example for
 - i. parallelogram of forces
 - ii. triangle of forces
 - iii. polygon of forces **[6]**
- i. Explain what is meant by simply supported beam **[2]**
- j. state the principle of moments **[2]**

SECTION B- answer any 5 questions

QUESTION 2

A vehicle having a mass of 750kg is being propelled along a level road by a force of 1.2kN against a total resistance to motion of 300N **[12]**

Calculate

- a. The acceleration of the vehicle
- b. How far the vehicle will travel from the rest in the first 15s
- c. The velocity of the vehicle at end of this time **[12]**

QUESTION 3

A flywheel of diameter 360mm increases its speed uniformly from 630 rev/min to 1050 rev/min in 11s

Calculate

- a. Angular acceleration of the wheel in rad/s^2
- b. Number of revolutions made during the speed change
- c. linear acceleration of a point on the rim of the wheel in m/s^2 **[12]**

QUESTION 4

In a drop forging operation the top die and its holder which have a combined mass of 50kg, fall freely on the bottom die.

- a. Kinetic energy of the top die just before striking the bottom die of its velocity at that instant is 7.8m/s
- b. The height through which the top die has fallen **[12]**

QUESTION 5

A point O is in equilibrium under the action of five coplanar forces as shown below. Find the magnitude of the two forces P and Q by construction Use scale $1\text{ cm} = 2\text{ N}$

QUESTION 6

A motor–lorry frame can be considered as a beam 4.2m long simply supported at two points, A and B, 3m apart. The frame is subjected to concentrated loads of 5kN, 10kN, 15kN and 10kN

Calculate

- a.** The value of support reactions at A and B
 - b.** The position at which the 15kN load must be applied to make the support reactions equal
- [12]**

QUESTION 7

A vehicle accelerated uniformly at 2.5 m/s^2 from 27 km/h

Calculate

- a. Time to attain a speed of 81 km/h
- b. Distance travelled in this time

If the vehicle is brought to rest with uniform retardation from 81 km/h in a further distance of 225 m , find

- c.i. retardation
- ii. time taken [12]

QUESTION 8

A motor car of mass $1\,000 \text{ kg}$ stands on an incline whose gradient is 1.8 . The handbrake is released and the vehicle runs down the incline. If frictional resistance to the motion is 40 N . Find the speed of the vehicle after running 200 m down the incline **[12]**